

Aryan Raut

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Experience

- Teaching Assistant** — *Rutgers University, New Brunswick, NJ* Sept 2025 – Current
◦ Leading a class of 50 students in CS344, Design and analysis of Algorithms, holding weekly in-person recitations and office hours.
- Student Tutor** — *Rutgers Learning Center, New Brunswick, NJ* Jan 2024 – Current
◦ Tutored 100+ students in various software/technology skills (AI, Python, Data Analytics, Numerical Analytics, R, C, DSA, etc.) via one-on-one appointments.
- Data Analyst** — *Enigma Technical Society, Manipal, KA* Sept 2022 – Aug 2023
◦ Collaborated with the data analytics team and used predictive analytics through Data Mining to cut event resource waste by 25% and expenses by 18%, improving decision-making speed by 40% with interactive Power BI dashboards.
- Web Developer** — *Medilance Healthcare, Chandigarh, PB* May 2022 – Aug 2022
◦ Worked in Web Dev team, enhanced performance, reducing page load times by 10-15% through lazy loading and optimization, while improving user engagement by reorganizing the product catalog for better navigation of a large scale website system.

Education

- Rutgers NB, State University of New Jersey** — *BSc Computer Science, Minor in Mathematics* Sept 2023 – Dec 2025
◦ **CGPA:** 3.8/4.0 — *Dean's List* in all semesters | *Magna Cum Laude*
◦ **Coursework:** AI, ML, Computer Security, Graphics, Artificial Neural Networks, OS, Data Structures and Algorithms.

Research Projects

- Comparative Analysis of Language Translation Accuracy** — 🌐
◦ Developed an evaluation framework comparing **GPT-3.5**, **Google Translate**, and **OPUS-MT** across language pairs, achieving **7% higher BLEU scores** with fine-tuned OPUS-MT for low-resource languages, and utilizing **TensorFlow** and **Hadoop** to cut training time by 35%.
- Credit Card Fraud Detection Model** — 🌐
◦ Engineered an **XGBoost-LSTM ensemble** achieving **99.2% fraud detection accuracy** on Kaggle's dataset, **improved F1-score from 0.82 to 0.91 using SMOTE**, built a PyTorch anomaly detection system (1000+ transactions/sec), and **reduced false positives by 40%** through feature engineering and threshold optimization.

Coding Projects

- Real Time Gesture Recognition System** — 🌐
◦ Trained a GPU powered (CoreML) PyTorch model on a custom dataset of labeled landmarks to use in a real-time gesture control system(OpenCV) and increased accuracy and **reduced inference time by 60%**.
◦ **Tools Used:** Python, OpenCV, MediaPipe, PyTorch, CoreML, scikit-learn
- Face and Digit classification using Classifiers and Neural Networks** — 🌐
◦ Implemented a robust vision pipeline using CNNs for digit and face recognition with **98.7% accuracy on MNIST digit classifier** and **95.3% accuracy on face recognition**.
◦ **Tools Used:** PyTorch, OpenCV, Scikit-learn, NumPy, CUDA for GPU acceleration
- AutoResearcher Pro** — 🌐
◦ Built an agentic AI research assistant that autonomously processes academic queries, integrating OpenAI GPT-4 for generation and sentence-transformers for RAG-based document summarization and insight extraction.
◦ **Tools Used:** Python, LangChain, OpenAI API, HuggingFace Transformers, Streamlit.
- E-Commerce Store using MERN** — 🌐
◦ Built a store that provides a seamless shopping experience with user authentication, product listings, cart management, order processing, and payment integration, utilizing a MongoDB and Express and Node to build REST APIs.
◦ **Tools Used:** MongoDB, Express.js, React, Node.js.

Skills

- **Programming:** Python, Java, R, MySQL, JavaScript, TypeScript, Git, C, C++, MATLAB, Assembly(basic).
- **ML/AI:** Neural Networks, Deep Learning, Computer Vision, Natural Language Processing, Large Language Models
- **ML Frameworks:** PyTorch, TensorFlow, scikit-learn, Keras
- **Web Dev:** React, HTML/CSS, MongoDB, NoSQL, WebGL, Express, NodeJS, NextJS
- **Math:** Probability, Statistics, Advanced Calculus, Optimization, Actuarial Math